

ARYAN DALAL

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EDUCATION

University of California, Los Angeles (UCLA)

Sep. 2024 – June 2028

B.S. Applied Mathematics & Computer Engineering

Coursework: Algorithms, Computer Systems, Convex Optimization, Numerical Methods, Linear Algebra, Probability Theory

Awards/Honors: \$2,000 Research Award — UCLA Undergraduate Research Center (URC) in Sciences

Activities: Association of Computing Machinery, IEEE, Undergraduate Math Student Association

TECHNICAL SKILLS

Languages / Systems: Python, C/C++, R, x86-64 Assembly, MIPS, UNIX/Linux

ML / AI: PyTorch, TensorFlow, Keras, HuggingFace, Sklearn, Weights & Biases, Optuna, MLflow

Tools / Profiling: Torchinfo, Ptflops, NVIDIA Nsight systems, NumPy, Pandas, Matplotlib, SciPy

Software Tools: Git, GNU GCC/Make, Shell Scripting, PostgreSQL, MongoDB, Docker, Pytest Debugging, Unit Testing

WORK EXPERIENCE

NASA

June 2026 – August 2026

Machine Learning Intern — AI Research & Model Development for C code Embedded Hardware

Cleveland, OH

- Implemented a hybrid CNN-ESN-Classifer pipeline for real-time anomaly detection on edge devices, achieving $F1 = 0.97$ on ECG5000 within a 68 KB memory footprint against a 192 KB hardware budget.
- Built a custom memory estimation and budget validation framework for embedded deployment, tracking static weight, peak activation and runtime memory constraints separately across each model to ensure hardware compliance prior to training.
- Theorized an Echo State Network inference pipeline targeting INT8 quantization for C code deployment on embedded hardware.
- Applied weighted binary cross-entropy with precision-recall optimization to address class imbalance for anomaly detection.
- Mentored by Brooke Weborg, Machine Learning Engineer, NASA Glenn Research Center, Advanced Computing Branch.

UCLA Mathematical Machine Learning Group

October 2025 – Present

Undergraduate Researcher — Quantification & Verification of ML/AI Systems

Los Angeles, CA

- Developed a PyTorch-based CNN pipeline for the SpuCoMNIST benchmark, using clustering-based group discovery and group-balanced ERM to mitigate spurious correlations and improve robustness to distribution shift.
- Used WizardLM Evol-Instruct method to augment a dataset generating 1200 labeled pairs for open-source Large Language Model fine-tuning with Quantized Low Rank Adoption (QLoRA) in Macaulay2 Computational Algebra programming language.
- Advised by Guido Montúfar and Yulia Alexandr in analytical approaches to neural network verification for safety-critical applications and algebraic techniques in statistical modeling and machine learning respectively.

TECHNICAL PROJECTS

Text Prediction from Neural Signals with Speech Brain-Computer Interface Decoding

Sept. 2025 - Dec. 2025

Brain-Computer Interface, Neural Signal Processing, Machine Learning

github.com/SpeechBCIDecoding

- Reduced Phoneme Error Rate (PER) of a baseline Gated-Recurrent Unit neural prediction model by 16.34% from 0.2268 to 0.1896 through architecture enhancements to a recurrent neural network model for Speech BCI decoding on Brain-to-text Benchmark '24 dataset
- Implemented Batch Layer Normalization to GRU inputs stabilizing training and convergence following a Transformer-based neural decoding architecture alongwith an AdamW Optimization to reduce overfitting and improve generalization with a small weight decay of 10^{-5} and a dropout rate of 0.35, increased from 0.2 to both GRU and post-GRU layers.

Skin Cancer Detection Model with 3D-Total Body Photographs

June 2025 - August 2025

Computer Vision, Image Processing, Ensemble Learning Methods

- Stacked ML algorithms such as SVM (87.90% accuracy), Classification and Regression Tree (86.80%), Light Gradient Boosting-Machine (87.50%) and XGBoost with Grid Search CV (90.70%) to improve f1-score and stacking accuracy.
- Ensemble Model correctly detected Benign cases with 93% precision and 94% recall with high support favoring overall benign performance over malignant cases.

VOLUNTEERING, ACTIVITIES, INTERPERSONAL

- ACM AI Workshops Officer: Taught fundamentals of Machine Learning, Computer Vision and Transformers to undergraduates
- UCLA Math Learning Assistant: Undergraduate TA for Calculus II and Honors Linear Algebra; Hosted proof-writing workshop
- Language Proficiency: English (Fluent), Hindi (Fluent), Spanish (Intermediate)